# KYG - Know Your Genes

Phase 1 Website Specification

Target Launch: May 14, 2026

Competitor Reference: mapmygenome.in (Shopify-based)

# 1. Why Next.js Over Shopify

## What Shopify Gives You

- Quick storefront setup
- Built-in payment gateway
- Basic product catalog and checkout
- Hosting included

## What Shopify Cannot Do (Why KYG Needs Next.js)

| Capability | Shopify | Next.js |
|---|---|---|
| User Dashboard | Not possible. No concept of a logged-in user dashboard with history, reports, preferences. | Full control. Build any authenticated experience - report history, kit tracking, health timelines, loyalty points. |
| Role-Based Access | Only store admin roles. No custom roles (user, counsellor, admin, lab-partner). | Implement any RBAC model with middleware-level route protection. |
| Shiprocket Integration (Forward + Reverse) | Requires third-party Shopify app (limited, paid, no reverse-pickup control). | Direct Shiprocket API integration. Full control over forward delivery of kit AND reverse pickup of sample. |
| Report Delivery System | No way to deliver genetic test reports inside the platform. You'd email PDFs separately. | Reports rendered in-app, tied to user account, with history and doctor/counsellor annotations. |
| Genetic Counsellor Booking | Not native. Need Calendly embed or external tool. | Built-in booking system tied to user profile and test results. |
| Quiz / Risk Assessment | Basic Shopify quiz apps exist but are shallow and don't integrate with your data. | Custom quiz logic that feeds into product recommendations and stores results in user profile. |
| Loyalty & Rewards | Third-party apps with monthly fees, limited customization. | Custom loyalty card system integrated with purchase history and referrals. |
| Blog with SEO Control | Basic blog, limited SEO flexibility. | Full MDX/CMS-driven blog with schema markup, OG tags, programmatic SEO. |
| Multi-stage User Journey | Shopify is transactional (buy and leave). | Lifecycle engagement: education -> purchase -> kit tracking -> sample -> report -> counselling -> follow-up. |
| Partner/B2B Portal | Not feasible. | Dedicated partner-with-us flow, B2B dashboards for clinics/hospitals. |
| Government Program Integration | Not possible. | Custom flows for subsidized testing, bulk orders, program-specific reporting. |
| Data Privacy & Compliance | Data lives on Shopify's servers (US/Canada). Genetic data has strict Indian regulations. | Self-hosted or India-region cloud. Full control over where genetic data is stored. |
| Cost at Scale | Shopify Plus starts at $2,000+/month. Transaction fees on every order. | Vercel/AWS hosting: fraction of the cost. No per-transaction platform fees. |

### The Core Argument

MapMyGenome uses Shopify because they treat genetic testing as a product you buy. KYG's value proposition is different - it's a health relationship platform where the test kit purchase is just the entry point. The real product is the ongoing dashboard, reports, counselling, literacy content, and lifecycle engagement. Shopify cannot be this platform. Next.js can.

## 2. Recommended Tech Stack (Phase 1)

| Layer | Technology | Reason |
|---|---|---|
| Framework | Next.js 15 (App Router) | SSR + SSG, API routes, middleware auth, ISR for blog/content |
| Language | TypeScript | Type safety across the full stack |
| Styling | Tailwind CSS + shadcn/ui | Rapid UI development, consistent design system |
| Auth | NextAuth.js (Auth.js v5) | Google/Email login, role-based sessions, JWT |
| Database | PostgreSQL (Supabase or Neon) | Relational data for users, orders, reports, roles |
| ORM | Prisma | Type-safe database queries |
| Payments | Razorpay | Indian payment gateway, UPI/cards/netbanking |
| Shipping | Shiprocket API | Forward + reverse logistics for kit delivery |
| CMS (Blog) | MDX files or Sanity CMS | Content team can publish without deploys |
| Email | Resend or AWS SES | Transactional emails (order confirmation, report ready) |
| Hosting | Vercel | Zero-config Next.js deployment, edge functions, India CDN |
| Storage | AWS S3 (Mumbai region) | Report PDFs, user documents - data stays in India |

## 3. Phase 1 - Complete Page Map

### 3.1 Marketing Pages (Public)

| # | Route | Purpose | Key Elements |
|---|---|---|---|
| 1 | / | Homepage | Hero, value props, featured packages, trust signals, CTA |
| 2 | /how-it-works | 4-step process | Order -> Receive Kit -> Collect Sample -> Get Report |
| 3 | /why-kyg | Differentiators | Indian-specific reference data, affordability, privacy-first, counsellor access |
| 4 | /science | Scientific credibility | Lab certifications, methodology, accuracy stats, research backing |
| 5 | /about | Company story | Mission, team, advisors, lab partners |
| 6 | /contact | Contact form + info | Form, email, phone, office address, Google Maps embed |
| 7 | /press | Media coverage | Press releases, media mentions, downloadable press kit |
| 8 | /quiz | Risk/recommendation quiz | Interactive quiz -> personalized package recommendation |
| 9 | /partner-with-us | B2B lead capture | For clinics, hospitals, corporate wellness, government programs |

## 3.2 Commerce Pages (Public + Auth)

| # | Route | Purpose | Key Elements |
|---|---|---|---|
| 1 | /packages | Product catalog | All test packages with pricing, comparison table |
| 2 | /packages/[slug] | Package detail | Description, biomarkers, sample report, FAQ, add-to-cart |
| 3 | /cart | Shopping cart | Line items, quantity, promo code, price summary |
| 4 | /checkout | Checkout flow | Address (Shiprocket pin check), payment (Razorpay), order summary |
| 5 | /checkout/success | Order confirmation | Order ID, expected delivery, next steps |
| 6 | /track/[orderId] | Order tracking | Shiprocket tracking - kit delivery + reverse pickup status |

## 3.3 Content Pages (Public)

| # | Route | Purpose | Key Elements |
|---|---|---|---|
| 1 | /blog | Blog listing | Categories: Genetic Literacy, Wellness, Research, Stories |
| 2 | /blog/[slug] | Blog post | Full article, author, share buttons, related posts, CTA |
| 3 | /faq | FAQ | Accordion-style, categorized (Testing, Privacy, Reports, Shipping) |
| 4 | /counsellor | Genetic counselling | What is counselling, counsellor profiles, booking CTA |

## 3.4 Legal Pages (Public)

| # | Route | Purpose |
|---|---|---|
| 1 | /privacy | Privacy policy - critical for genetic data trust |
| 2 | /terms | Terms of service |
| 3 | /cookies | Cookie policy |

## 3.5 Auth Pages

| # | Route | Purpose | Key Elements |
|---|---|---|---|
| 1 | /login | User login | Email/password, Google OAuth, OTP option |
| 2 | /signup | User registration | Name, email, phone, password, terms acceptance |
| 3 | /forgot-password | Password reset | Email input -> reset link -> new password |

## 3.6 User Dashboard (Authenticated)

| # | Route | Purpose | Key Elements |
|---|---|---|---|
| 1 | /dashboard | Overview | Active orders, recent reports, upcoming counselling, loyalty points |
| 2 | /dashboard/orders | Order history | All orders with status, tracking links, reorder option |
| 3 | /dashboard/orders/[id] | Order detail | Full order info, Shiprocket tracking, kit instructions |
| 4 | /dashboard/reports | Report history | All genetic test reports, download PDF, view in-app |
| 5 | /dashboard/reports/[id] | Individual report | Full report with explanations, actionable insights |
| 6 | /dashboard/profile | Profile management | Personal info, health profile, communication preferences |
| 7 | /dashboard/family | Family history | Family tree builder, hereditary risk flags |
| 8 | /dashboard/loyalty | Loyalty / rewards | Points balance, earn/redeem history, referral link |
| 9 | /dashboard/bookings | Counsellor appointments | Book, reschedule, cancel counselling sessions |

### 3.7 Admin Panel (Role: Admin)

| # | Route | Purpose |
|---|---|---|
| 1 | /admin | Admin dashboard - order stats, revenue, user counts |
| 2 | /admin/orders | Manage all orders, update status, trigger Shiprocket actions |
| 3 | /admin/users | User management, role assignment |
| 4 | /admin/reports | Upload/manage genetic reports for users |
| 5 | /admin/content | Blog post management |
| 6 | /admin/packages | Manage test packages, pricing, availability |
| 7 | /admin/coupons | Promo codes and discount management |

## 4. Shiprocket Integration Flow

### Forward Flow (Kit to Customer)

- Order Placed
- Create Shiprocket Order (POST /orders/create)
- Generate AWB
- Kit Shipped
- Customer Receives Kit
- Status: 'Kit Delivered'

### Reverse Flow (Sample back to Lab)

- Customer Collects Sample
- Request Reverse Pickup via Shiprocket API (POST /orders/create/return)
- Pickup Scheduled
- Sample Collected
- Sample Reaches Lab
- Status: 'Sample Received'
- Lab Processing (3-4 weeks)
- Report Ready -> Notify User

### Key Shiprocket APIs

- POST /orders/create - forward shipment
- POST /orders/create/return - reverse pickup
- GET /courier/track/shipment/{id} - tracking
- GET /courier/serviceability - pin code check at checkout

## 5. User Roles

| Role | Access |
|---|---|
| Guest | Marketing pages, blog, FAQ, quiz, package browsing |
| User | Everything above + dashboard, orders, reports, bookings, loyalty |
| Counsellor | User access + counsellor dashboard, patient report access, session management |
| Admin | Full access including admin panel, order management, report upload, user management |

## 6. Phase 1 Priority (MVP for May 14, 2026 Launch)

### Must Have (Launch Blockers)

- Homepage with clear value proposition
- How It Works page
- Package catalog + detail pages (minimum 3 packages)
- Cart + Checkout with Razorpay
- Shiprocket forward delivery integration
- User signup/login (email + Google)
- User dashboard with order tracking
- Basic blog (5-10 genetic literacy articles for SEO)
- Contact page
- Privacy policy + Terms
- Mobile responsive design
- Admin: order management + report upload

### Should Have (Week 1-2 Post-Launch)

- Quiz / recommendation engine
- Shiprocket reverse pickup integration
- Report viewing in dashboard
- FAQ page
- Coupon/promo code system

### Nice to Have (Phase 2)

- Loyalty card system
- Counsellor booking
- Family history builder
- Partner portal
- Service add-ons (Cult Fit, Paytm tie-ups)
- Age-by-stages journey mapping
- Government program flows
- NRI-specific packages and shipping

## 7. Target User Segments

| Segment | Key Need | Entry Point |
|---|---|---|
| Pre-Conception Couples | Carrier screening, hereditary risk | Quiz, Why KYG, packages |
| Family History Concerns | Cancer/cardiac/diabetes risk assessment | Science page, counsellor |
| Chronic Care Patients | Pharmacogenomics, treatment optimization | Doctor referral, partner portal |
| NRI Segment | Indian-specific reference data, family testing | Direct marketing, premium packages |
| Bio-Hackers / Young Parents | Wellness optimization, proactive health | Blog, quiz, social media |

## 8. Key Differentiators to Highlight on Website

- Indian-Specific Reference Data - Not Western averages applied to Indian genetics
- Affordable - Lower barrier to entry, EMI options, government program alignment
- Privacy-First - Data stored in India, user controls data, no third-party sharing
- Actionable Reports - Not just data dumps, clear next steps with counsellor support
- Genetic Literacy - Education-first approach builds trust before selling
- No Marriage Stigma - Sensitive handling of carrier status, counselling-first approach
- Clinical Grade - Not just ancestry/wellness fluff, actual clinical utility

## 9. What we Need from the Account Manager (Teesha)

Every item below must be provided, confirmed, or decided by the business/account team BEFORE development can begin on that module. Items tied to a milestone are blocking — dev cannot proceed past that milestone without them.

### 9.1 Detailed Requirement List

### A. Brand & Design Assets

| # | Requirement | Details | Status |
|---|---|---|---|
| 1 | Brand guidelines document | Logo (SVG + PNG), primary/secondary colors (hex), typography (fonts + weights), tone of voice | [ ] |
| 2 | Favicon and app icons | 16x16, 32x32, 180x180, 512x512 PNG + SVG | [ ] |
| 3 | Hero images / lifestyle photography | Homepage hero, How It Works visuals, About page team photos | [ ] |
| 4 | Product images | Photos of each test kit (box, contents, swab/tube), lifestyle shots | [ ] |
| 5 | UI/UX wireframes or design mockups | Figma/Adobe XD files for key pages (homepage, package detail, dashboard, checkout) | [ ] |
| 6 | Social media handles & links | Instagram, LinkedIn, Twitter/X, YouTube, Facebook | [ ] |

## B. Product & Commerce Data

| # | Requirement | Details | Status |
|---|---|---|---|
| 7 | Complete product catalog | For EACH package: name, SKU, price (MRP + selling), short/long description, biomarkers list, turnaround time, sample type | [ ] |
| 8 | Package comparison matrix | Which features/tests are in which package | [ ] |
| 9 | Sample report PDFs | At least one anonymized sample report per package | [ ] |
| 10 | Pricing strategy | Launch discounts, bundle pricing, EMI options, pre-configured coupon codes | [ ] |
| 11 | Kit contents list | Exact items in each kit box | [ ] |
| 12 | Sample collection instructions | Step-by-step guide with images for customer sample collection | [ ] |

## C. Shipping & Logistics

| # | Requirement | Details | Status |
|---|---|---|---|
| 13 | Shiprocket account credentials | API email, password, and channel ID | [ ] |
| 14 | Pickup address(es) | Warehouse/lab address for forward shipment origin | [ ] |
| 15 | Return/reverse pickup address | Lab address where samples should be shipped back to | [ ] |
| 16 | Shipping pricing decision | Free shipping? Flat rate? Threshold-based (free above X)? | [ ] |
| 17 | Serviceable pin codes | Any restrictions on which areas you can serve, or all-India? | [ ] |
| 18 | Kit weight and dimensions | For accurate Shiprocket rate calculation | [ ] |
| 19 | Packaging requirements | Any cold-chain/temperature requirements for reverse shipment? | [ ] |

## D. Payments & Legal

| # | Requirement | Details | Status |
|---|---|---|---|
| 20 | Razorpay account credentials | Key ID + Key Secret (test mode first, then live) | [ ] |
| 21 | GST number and details | For invoice generation | [ ] |
| 22 | Business legal name | For receipts, invoices, terms of service | [ ] |
| 23 | Refund/cancellation policy | Clear policy — conditions, timelines, partial/full | [ ] |
| 24 | Privacy policy content | Lawyer-reviewed privacy policy (especially for genetic data) | [ ] |
| 25 | Terms of service content | Lawyer-reviewed ToS text | [ ] |
| 26 | Cookie policy content | Lawyer-reviewed cookie policy | [ ] |

## E. Content & Copy

| # | Requirement | Details | Status |
|---|---|---|---|
| 27 | Homepage copy | Headline, subheadline, value propositions, CTAs | [ ] |
| 28 | About page content | Company story, mission/vision, founder bios + photos, advisor bios | [ ] |
| 29 | How It Works copy | Step-by-step text and any custom illustrations | [ ] |
| 30 | Why KYG copy | Differentiator messaging, proof points, statistics | [ ] |
| 31 | Science page content | Lab certifications, accreditations, methodology, research references | [ ] |
| 32 | FAQ content | Minimum 20 Q&A pairs across categories | [ ] |
| 33 | Blog articles | Minimum 5-10 articles for launch (Genetic Literacy focus) | [ ] |
| 34 | Counsellor page content | Counsellor profiles, photos, qualifications | [ ] |
| 35 | Press page content | Existing press coverage, press release drafts, media kit | [ ] |
| 36 | Contact information | Support email, phone, office address, support hours | [ ] |
| 37 | Quiz questions and logic | Questions, answer options, scoring logic, package mapping | [ ] |

**F. Third-Party Accounts & Integrations**

| # | Requirement | Details | Status |
|---|---|---|---|
| 38 | Domain name | Confirmed domain (e.g., knowyourgenes.in / kyg.in) + DNS access | [ ] |
| 39 | Google Analytics / Tag Manager | GA4 measurement ID, GTM container ID | [ ] |
| 40 | Google OAuth credentials | Google Cloud Console project + OAuth client ID | [ ] |
| 41 | Email service credentials | Resend API key or AWS SES setup | [ ] |
| 42 | Cloud hosting account | Vercel team or AWS account for deployment | [ ] |
| 43 | Database hosting | Supabase project or Neon DB instance | [ ] |
| 44 | S3 bucket | AWS S3 bucket (Mumbai region) for report storage | [ ] |
| 45 | Meta Pixel / ad tracking IDs | Facebook Pixel, Google Ads conversion tag | [ ] |
| 46 | WhatsApp Business API | If WhatsApp support/notifications needed at launch | [ ] |

**G. Business Logic Decisions**

| # | Requirement | Details | Status |
|---|---|---|---|
| 47 | User roles confirmation | Confirm: Guest, User, Counsellor, Admin — any others? | [ ] |
| 48 | Report delivery workflow | Who uploads reports? Manual admin upload or lab system integration? | [ ] |
| 49 | Counsellor booking system | Manual or self-service? Video call platform? | [ ] |
| 50 | Loyalty points structure | How are points earned? Redemption value? Referral bonus? | [ ] |
| 51 | Partner program details | Commission structure, partner tiers, approval process | [ ] |
| 52 | Email notification list | Which events trigger emails? | [ ] |
| 53 | SMS/WhatsApp notification list | Which events need SMS/WhatsApp? | [ ] |
| 54 | Age restrictions | Minimum age for ordering? Parental consent? | [ ] |

## 9.2 Comprehensive Development Checklist

### Phase 1A: Foundation (Week 1 — Apr 17 to Apr 23)

- Project setup — Next.js 15, TypeScript, Tailwind, ESLint, Prettier
- Design system — shadcn/ui components, brand colors, typography configured
- Database schema — Prisma models for Users, Orders, Products, Reports, Roles
- Auth system — NextAuth.js with email/password + Google OAuth + role middleware
- Layout components — Header, Footer, Mobile Nav, Dashboard Sidebar
- SEO foundation — Metadata, sitemap, robots.txt, OG image template

**Blocking dependencies: Items #1, #2, #6, #38, #40, #42, #43, #47**

### Phase 1B: Marketing Pages (Week 2 — Apr 24 to Apr 30)

- Homepage — Hero, packages preview, how-it-works summary, trust signals, CTAs
- How It Works — 4-step visual timeline
- Why KYG — Differentiators section
- Science — Credibility content
- About — Team, mission, story
- Contact — Form + Google Maps + info

- Press — Media coverage listing
- Privacy / Terms / Cookies — Legal pages

**Blocking dependencies: Items #3, #24, #25, #26, #27, #28, #29, #30, #31, #35, #36**

### Phase 1C: Commerce (Week 2-3 — Apr 28 to May 5)

- Product catalog page — Grid/list view with filters
- Product detail page — Full package info, add-to-cart, sample report
- Cart system — Add/remove/update, promo code, persistent cart
- Checkout flow — Address form, Shiprocket pin check, Razorpay integration
- Order confirmation page — Success state with next steps
- Order tracking page — Shiprocket tracking integration
- Shiprocket forward delivery — API integration for order creation + AWB
- Razorpay payment gateway — Test + live mode, webhooks

**Blocking dependencies: Items #7, #8, #9, #10, #11, #12, #13, #14, #16, #17, #18, #20, #21, #22, #23**

### Phase 1D: User Dashboard (Week 3 — May 1 to May 7)

- Dashboard overview — Active orders, recent activity
- Order history — List with status badges, tracking links
- Order detail — Full info + Shiprocket tracking embed
- Reports section — List + individual report view (PDF download)
- Profile management — Edit personal info, change password
- Email notifications — Order confirmed, shipped, delivered, report ready

**Blocking dependencies: Items #41, #44, #48, #52**

### Phase 1E: Content & Admin (Week 3-4 — May 5 to May 10)

- Blog system — Listing + individual posts + categories + search
- FAQ page — Accordion by category
- Counsellor info page — Profiles, booking CTA
- Quiz engine — Interactive quiz with package recommendation
- Admin: Dashboard — Order stats, user counts, revenue
- Admin: Order management — View/update orders, trigger Shiprocket
- Admin: Report upload — Upload PDF reports linked to user + order
- Admin: User management — View users, assign roles
- Admin: Package management — CRUD for test packages
- Admin: Coupon management — Create/edit promo codes

**Blocking dependencies: Items #32, #33, #34, #37**

### Phase 1F: QA & Launch (Week 4 — May 10 to May 14)

- Cross-browser testing — Chrome, Safari, Firefox, Edge
- Mobile responsive QA — iOS Safari, Android Chrome
- Payment testing — Full Razorpay flow with test cards
- Shipping testing — Shiprocket order creation + tracking in test mode
- Performance audit — Lighthouse score > 90, Core Web Vitals pass
- SEO audit — All meta tags, OG images, structured data, sitemap
- Security review — Auth flows, API route protection, input sanitization, HTTPS
- Analytics setup — GA4 + events for key actions
- Error monitoring — Sentry or similar for production error tracking
- DNS + domain setup — Point domain to Vercel, SSL certificate
- Go-live deployment — Production build, environment variables, final smoke test

**Blocking dependencies: Items #39, #45, #46 (if applicable)**

## 9.3 Milestones & Timeline

**TARGET LAUNCH: MAY 14, 2026 | TOTAL DEVELOPMENT TIME: 4 WEEKS**

| Milestone | Date | Deliverable | Go/No-Go Criteria |
|---|---|---|---|
| M0: Kickoff | Apr 17 (Thu) | Spec document signed off | Account manager has reviewed and approved scope |
| M1: Foundation Ready | Apr 23 (Wed) | Auth working, DB live, layout done, deployed to staging | Can sign up, log in, see dashboard shell. Requires: Brand assets, domain, hosting, Google OAuth |
| M2: Marketing Site Live | Apr 30 (Wed) | All 8 marketing + 3 legal pages on staging | Content-complete and mobile responsive. Requires: All page copy, images, legal documents |
| M3: Commerce Working | May 5 (Mon) | Full buy flow on staging | Can place test order, payment works, Shiprocket order created. Requires: Catalog, Razorpay + Shiprocket creds |
| M4: Dashboard + Admin | May 8 (Thu) | User dashboard + admin panel on staging | User sees orders/reports. Admin manages orders + uploads reports. Requires: Report workflow, email service |
| M5: Content Complete | May 10 (Sat) | Blog, FAQ, quiz, counsellor pages live | All content populated with real content. Requires: Articles, FAQ, quiz logic, counsellor profiles |
| M6: QA Complete | May 12 (Mon) | All testing done, bugs fixed | Lighthouse > 90, no critical bugs, payment + shipping tested e2e |
| M7: LAUNCH | May 14 (Wed) | Production deployment, domain pointed, go-live | Final smoke test passed, analytics firing, monitoring active |

## 9.4 Critical Path — What Blocks Everything

These items, if delayed, push the entire launch date:

| Priority | Item | Needed By | Impact if Delayed |
|---|---|---|---|
| P0 | Brand guidelines + logo (#1) | Apr 17 | Blocks all UI work |
| P0 | Domain name + DNS access (#38) | Apr 17 | Blocks deployment + SSL |
| P0 | Razorpay credentials (#20) | Apr 25 | Blocks checkout — 2 week lead time for live activation |
| P0 | Shiprocket credentials (#13) | Apr 25 | Blocks order fulfillment flow |
| P0 | Product catalog data (#7) | Apr 25 | Blocks all commerce pages |
| P1 | Homepage + marketing copy (#27-31) | Apr 25 | Blocks marketing site milestone |
| P1 | Legal documents (#24, #25, #26) | Apr 28 | Blocks legal pages — needs lawyer review time |
| P1 | Google OAuth credentials (#40) | Apr 20 | Blocks social login |
| P2 | Blog articles (#33) | May 3 | Blocks content milestone but not launch |
| P2 | Quiz logic (#37) | May 3 | Can launch without quiz if needed |

## 9.5 Communication Protocol

| What | How | When |
|---|---|---|
| Daily standup | WhatsApp/Slack message | Every morning — what's done, what's blocked |
| Milestone review | 30-min call + staging demo | At each milestone date |
| Blocker escalation | Immediate call/message | When a dependency is overdue and blocking dev |
| Content handoff | Google Drive shared folder | Account manager uploads, developer pulls |
| Design handoff | Figma link with dev mode | Before each UI milestone |
| Bug reports | Shared sheet or Linear board | During QA phase (May 10-12) |